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Curso: Metodos numericos

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c = 17

95 | 70

$$f(x) = (\cos(cx)) - e^x$$

$$x_0 = 0 \quad , \quad x_1 = \frac{\pi}{3}$$

$$\int_0^{\frac{\pi}{3}} (\cos(17x)) - e^x \approx h\left(\frac{y_0 + y_1}{2}\right)$$

$$h = \frac{\pi}{3} - 0 = \frac{\pi}{3}$$

$$y_0 = f(x_0) = f(0) = (\cos(17 \times (0))) - e^0 = 1 - 1 = 0$$

$$y_1 = f(x_1) = f\left(\frac{\pi}{3}\right) = \left(\cos\left(17\,\frac{\pi}{3}\right)\right) - e^{\frac{\pi}{3}} = -2.34965$$

$$\int_0^{\frac{\pi}{3}} (\cos(17x)) - e^x \approx \frac{\pi}{3}\left(\frac{0 + (-2.34965)}{2}\right) = -1.23027 \rightarrow \text{Vaproximado}$$

Aplico COI

$$\int_0^{\frac{\pi}{3}} (\cos(17x) - e^x = \frac{1}{17}\operatorname{Sen}(17x) - e^x \Big|_0^{\frac{\pi}{3}} = \frac{1}{17}\operatorname{Sen}(17 \times 0) - e^0 - \frac{1}{17}\operatorname{Sen}\left(17 \times \frac{\pi}{3}\right) - e^{\frac{\pi}{3}} = -1.90059$$

$$= -1.90059 \rightarrow (\text{Vreal})$$

$$\text{Ea} = |-1.90059 - (-1.23027)| = 0.67032$$

Mi Nota
Tu marca genial

Escaneado con CamScanner